

Predictors of sPCI-pPCI Discordance: OLS and Robust Regression Analysis

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Sources: notebooks/02-assoc_difference_ols.ipynb, notebooks/02-assoc_difference_ols_r

1. Background and Objective

A systematic positive bias between intraoperative surgeon PCI (sPCI) and preoperative/pathological PCI (pPCI) has been established (mean sPCI – pPCI $\approx$ 6.9, see report 01-agreement_all_report.md). This analysis investigates **which patient and clinical characteristics explain the magnitude of this discordance**, using ordinary least squares (OLS) regression.

2. Study Sample

After excluding tumour type 8 and restricting to CC 0/1, the analytic sample comprised **N = 409** patients.

Tumour type 1 (n = 122) was set as the reference category.

Predictors included:

Variable	Type	Notes
Sex	Binary	0 = female, 1 = male
Age	Continuous	Years
CC	Binary	0 = complete cytoreduction, 1 = incomplete
PreOP CTx	Binary	Any preoperative chemotherapy (≥ 1 cycle)
Thermoablation	Binary	Intraoperative thermoablation performed
Tumour type	Nominal	Types 2-7 vs. reference type 1

3. Outcome Variables

Two outcomes were constructed:

- **raw_diff** = sPCI – pPCI (signed discordance)
- **abs_diff** = |sPCI – pPCI| (magnitude of discordance)

Statistic	raw_diff	abs_diff
Mean	7.09	7.39
SD	7.01	6.69
Median	5.0	5.0
Min	–8	0
Max	32	32

The positive mean and median of raw_diff confirm that surgeons consistently find **more** peritoneal disease intraoperatively than predicted preoperatively (median

discordance of 5 PCI points). Both distributions depart significantly from normality (Shapiro-Wilk: $p < 0.001$ for both).

4. OLS Regression Results (standard)

Outcome: raw_diff (sPCI – pPCI)

	F-statistic	df	p-value
Overall model	21.48	(11, 397)	3.0×10^{-34}

Variance explained: $R^2 = 0.373$, Adjusted $R^2 = 0.356$

The model is highly significant overall and explains approximately **37% of the variance** in discordance.

Regression Coefficients

Predictor	Coefficient	SE	t	p	95% CI
Intercept	+6.23	1.69	3.69	0.0003	[2.90, 9.55]
Sex (male)	−1.15	0.64	−1.79	0.074	[−2.40, 0.11]
Age	−0.01	0.02	−0.57	0.568	[−0.06, 0.03]
CC (in-complete)	+6.02	0.61	9.83	< 0.001	[4.81, 7.22]
PreOP CTx	−0.85	0.75	−1.13	0.260	[−2.32, 0.63]
Thermoablation	−1.39	0.70	−1.99	0.047	[−2.76, −0.02]
Tumour type 2	+0.05	1.36	0.04	0.972	[−2.63, 2.73]
Tumour type 3	+1.17	1.21	0.97	0.335	[−1.21, 3.54]
Tumour type 4	+1.64	0.92	1.78	0.076	[−0.17, 3.45]
Tumour type 5	+6.79	0.82	8.23	< 0.001	[5.17, 8.41]
Tumour type 6	−0.48	1.05	−0.46	0.647	[−2.54, 1.58]
Tumour type 7	+4.45	1.23	3.61	0.0003	[2.03, 6.87]

Key Findings

- **CC (incomplete cytoreduction)** is the strongest predictor: patients with incomplete cytoreduction (CC1) show on average **+6.0 points** more discordance than those with complete resection (CC0, $p < 0.001$).
- **Tumour type 5** shows **+6.8 points** more discordance than type 1 ($p < 0.001$).

- **Tumour type 7** shows **+4.5 points** more discordance than type 1 ($p < 0.001$).
- **Thermoablation** is associated with **1.4 points less** discordance ($p = 0.047$), possibly reflecting more localised and predictable disease in patients selected for this procedure.
- Sex, Age, PreOP CTx, and tumour types 2, 3, 4, and 6 are **not significant** predictors.

5. Robust Regression Results (Huber M-estimator)

Given the OLS violations (non-normality, heteroskedasticity), the model was re-estimated using an **Iteratively Reweighted Least Squares (IRLS) Huber M-estimator** (default HuberT, tuning constant 1.345), which downweights observations with large residuals. Bootstrap confidence intervals (2 000 resamples, seed 42) are reported alongside asymptotic standard errors.

Outcome: raw_diff (sPCI – pPCI)

Weighted R²: 0.383 | Scale (MAD): 5.06 | Durbin-Watson: 1.99

RLM Coefficients

Predictor	Coefficient	Asym. SE	p	95% CI (bootstrap)
Intercept	+7.08	1.65	< 0.001	[+3.77, +10.55]
Sex (male)	−0.82	0.63	0.191	[−2.06, +0.40]
Age	−0.02	0.02	0.329	[−0.07, +0.02]
CC (incomplete)	+5.53	0.60	< 0.001	[+4.08, +7.15]
PreOP CTx	−1.15	0.74	0.119	[−2.58, +0.22]
Thermoablation	−1.62	0.68	0.018	[−3.13, −0.08]
Tumour type 2	−0.05	1.33	0.970	[−2.28, +2.35]
Tumour type 3	+0.56	1.18	0.639	[−1.84, +3.18]
Tumour type 4	+1.25	0.90	0.166	[−0.33, +2.93]
Tumour type 5	+5.88	0.81	< 0.001	[+4.10, +7.98]
Tumour type 6	−0.92	1.03	0.370	[−2.57, +0.73]
Tumour type 7	+4.18	1.21	< 0.001	[+1.74, +6.83]

Key Findings (RLM)

- Results are consistent with OLS in direction and significance for all predictors.
- **CC (incomplete)** remains the strongest predictor: +5.5 points discordance vs. OLS +6.0; bootstrap CI excludes zero.
- **Tumour types 5 and 7** retain strong significance (+5.9 and +4.2 points respectively).
- **Thermoablation** −1.6 points ($p = 0.018$); bootstrap CI just excludes zero $[-3.13, -0.08]$.
- Sex, Age, PreOP CTx no longer approach significance under robust estimation.
- 20 high-leverage observations ($h > 2p/n = 0.059$); Cook's D never exceeds 1 ($\max = 0.042$), indicating no individually dominant observations.

6. Residual Diagnostics (OLS)

Test	Statistic	p-value	Conclusion
Shapiro-Wilk (residuals)	W = 0.980	< 0.001	Residuals non-normal
Jarque-Bera	20.16	< 0.001	Confirms non-normality
Breusch-Pagan (LM)	64.88	< 0.001	Heteroskedasticity present
Breusch-Pagan (F)	6.80	< 0.001	Heteroskedasticity present
Skewness	0.48	—	Mild right skew
Kurtosis	3.54	—	Slightly leptokurtic

Both OLS assumptions of **normality** and **homoskedasticity** of residuals are violated. The reported standard errors and p-values should be interpreted with caution; they are likely anti-conservative (i.e., p-values may be overstated).

7. Limitations and Recommendations

1. ~~Heteroskedasticity-robust standard errors (HC3)~~ **Huber M-estimator with bootstrap CIs** has been applied (see Section 5) and yields the preferred inference.
2. **Quantile regression** on the median would be more robust given the skewed, heteroskedastic outcome and would not rely on distributional assumptions.
3. The signed outcome (`raw_diff`) assumes direction matters. If only magnitude is of interest, rerunning the model on `abs_diff` should be considered.
4. Tumour types with small cell counts (types 2, 6, 7) warrant cautious interpretation due to limited statistical power.
5. The negative coefficient for **Thermoablation** likely reflects **selection bias** rather than a causal effect — patients undergoing thermoablation may have

more resectable, well-demarcated disease that is easier to assess preoperatively.

8. Summary

Both OLS ($R^2 = 0.37$) and Huber robust regression (weighted $R^2 = 0.38$) yield consistent conclusions across all predictors. **CC status** is the dominant driver ($\sim +5.5$ to $+6.0$ points for incomplete vs. complete cytoreduction). **Tumour types 5 and 7** carry substantially more intraoperative surprise than type 1. **Thermoablation** is associated with slightly less discordance (-1.4 to -1.6 points). Sex, age, and preoperative chemotherapy are not independently significant. The robust model confirms these findings are not driven by outliers (max Cook's $D = 0.042$, Durbin-Watson ≈ 2.0). OLS assumption violations (non-normality, heteroskedasticity) do not change the substantive interpretation, but bootstrap confidence intervals from the robust model are preferred for inference.